Name____Fernando Sotelo_____ Mark _____/50

1. Brief introduction __/3

My feature is called Fish Catalogue & Achievements. In this feature, whenever the player catches a fish or they meet a set of requirements, they gain an achievement. Additionally, when they achieve an achievement, they have a chance of gaining special items.

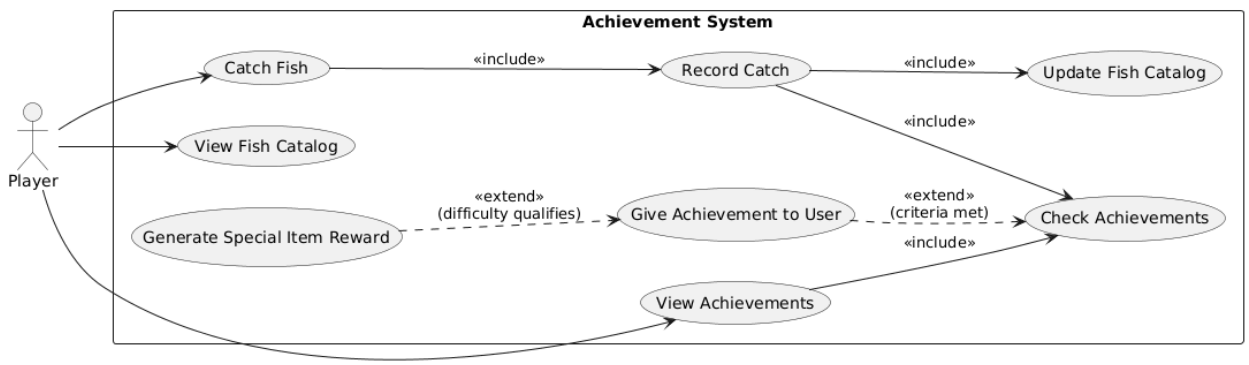
2. Use case diagram with scenario __14

[Use the lecture notes in class.

Ensure you have at least one exception case, and that the <<extend>> matches up with the Exceptions in your scenario, and the Exception step matches your Basic Sequence step.

Also include an <<include>> that is a suitable candidate for dynamic binding]

Use Case Diagrams



Scenarios

Name: Fish Catalogue & Achievements

Summary: A player catches a fish and wants the game to record it, update their Fish Catalogue, and award/check achievements (with a chance to receive a special item reward).

Actors: Player

Preconditions:

- The player is in gameplay and performs a valid fishing action **OR**
- The player opens the menu to view Fish Catalogue / Achievements

Basic sequence

Step 1: Player catches a fish.
Step 2: The system validates the catch data, then records the catch and updates the Fish Catalogue.
Step 3: The system checks whether the catch satisfies any achievement criteria.
Step 4: The player presses a button to view Fish Catalogue and/or Achievements, and the system retrieves and displays the updated data.

Extensions / Exceptions (to match the <<extend>> and boundary rules)

Step 3a (Extension – Award Achievement): If achievement criteria is met, the system awards the achievement and updates the achievement data.

Step 3a1 (Extension – Generate Reward): If the awarded achievement difficulty qualifies, the system randomly generates a special item reward and grants it to the player.

Step 2a (Exception – Invalid Catch Data): If the catch data is invalid (unknown fish, missing fields, or impossible values), the system displays an error and does not record the catch or update the Fish Catalogue.

Step 4a (Exception – No Saved Data): If the player has no Fish Catalogue entries or no Achievements yet, the system displays an empty/"none yet" state instead of failing.

Post conditions

- The Fish Catalogue is updated (if the catch was valid).
- Achievements are updated (if any were earned).
- The Fish Catalogue and Achievements are displayed when requested.
- A reward item is granted and shown if generated.

Priority: 2 (Essential)

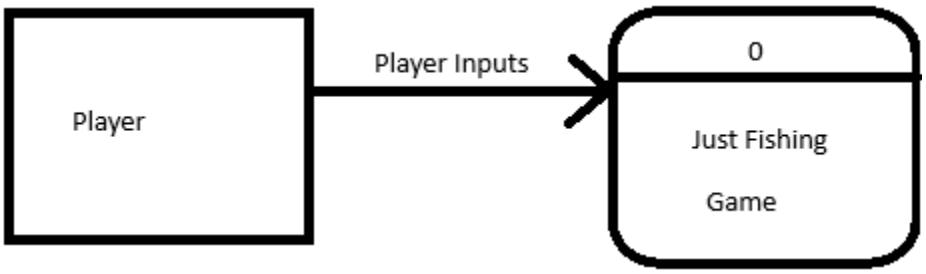
ID: C01

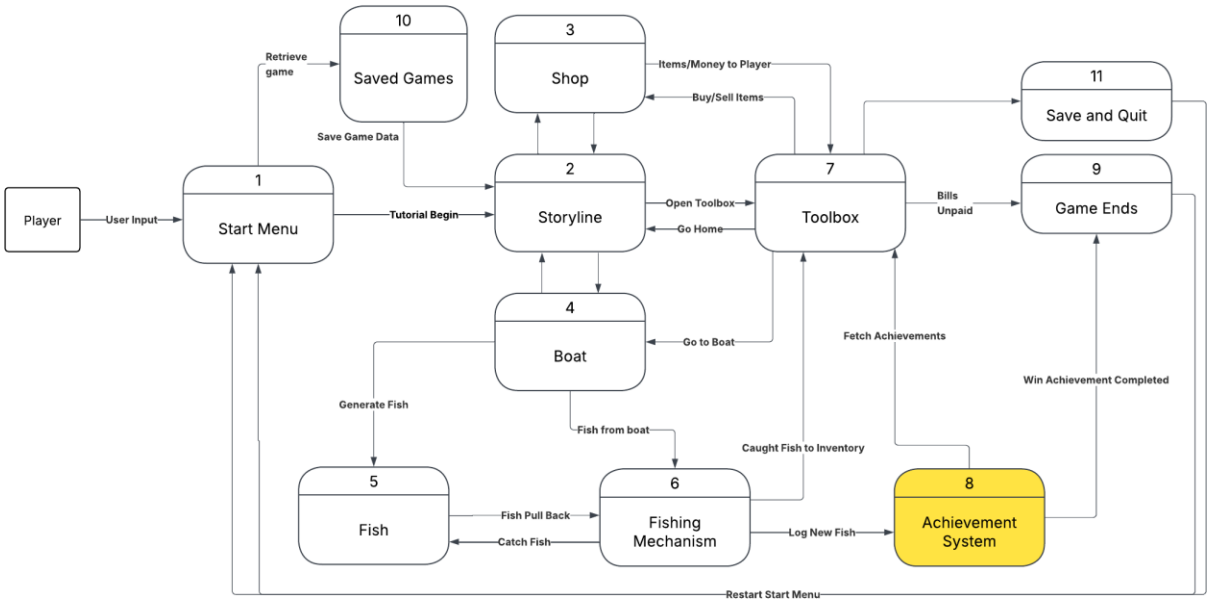
3. Data Flow diagram(s) from Level 0 to process description for your feature _____14

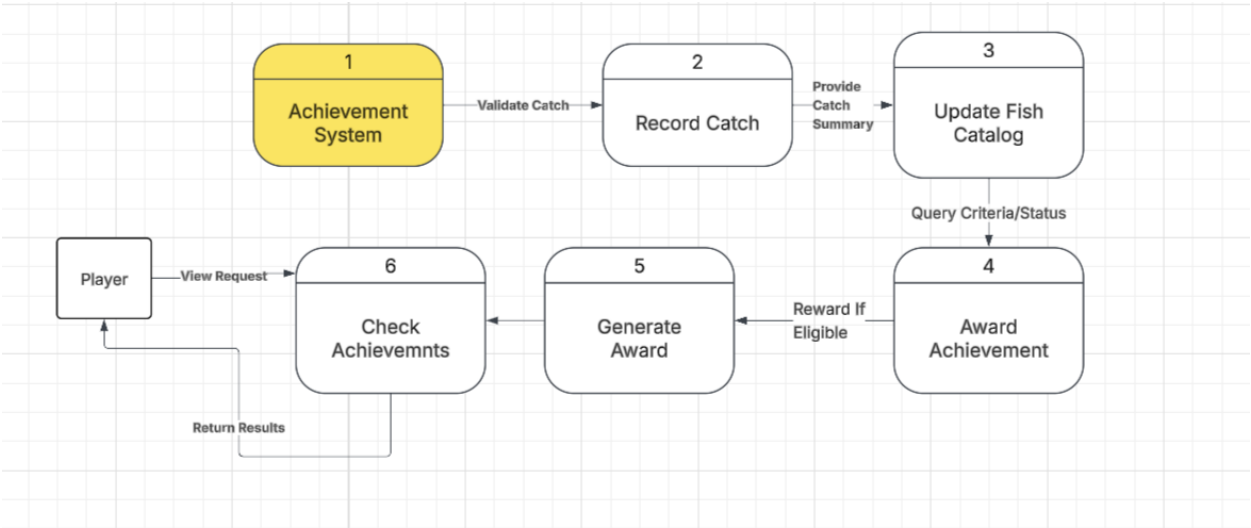
[Get the Level 0 from your team. Highlight the path to your feature]

Example:

Data Flow Diagrams







Process Descriptions

```
INPUT: fish_id, fish_length_cm

Step 1: Receive catch event from player

Step 2: Validate catch data
  IF fish_id is invalid OR missing THEN
     Display error message
     STOP process
  END IF

  IF fish_length_cm < 12 OR fish_length_cm > 40 THEN
     Display error message
     STOP process
  END IF

Step 3: Record catch
  Add fish to inventory
  Update Fish Catalogue data

Step 4: Evaluate achievements
  FOR each achievement
    IF achievement criteria met AND not already earned THEN
      Award achievement
      Update achievement data

      IF difficulty qualifies THEN
        Generate reward item
```

```
            Add reward to inventory
        END IF
    END IF
END FOR

Step 5: Allow user to view data
    IF player selects Catalogue or Achievements THEN
        Retrieve and display data
    END IF
```

4. Acceptance Tests _____9

[Describe the inputs and outputs of the tests you will run. Ensure you cover all the boundary cases.]

Achievement Criteria Validation

Output	Total Catches After Recording	Difficulty	Notes
No Achievement Awarded	9	Easy	Below threshold. Achievement requires 10 catches.
Achievement Awarded + Data Updated	10	Easy	Threshold met. Achievement awarded. No reward generated for Easy difficulty.
Achievement Awarded + Data Updated + Reward Generated	10	Hard	Threshold met. Hard difficulty qualifies for reward generation.
No New Achievement Awarded	11	Easy	Above threshold. Achievement should not duplicate once already earned.
No New Achievement Awarded	11	Hard	Above threshold. No duplicate achievement or additional reward.

Fish Length Validation

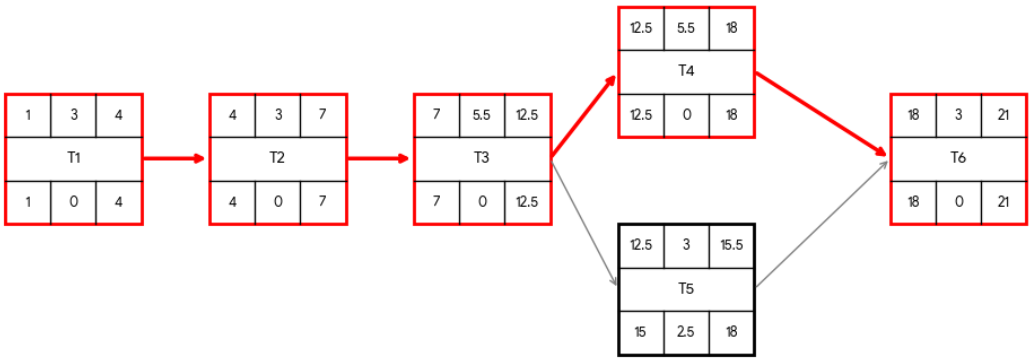
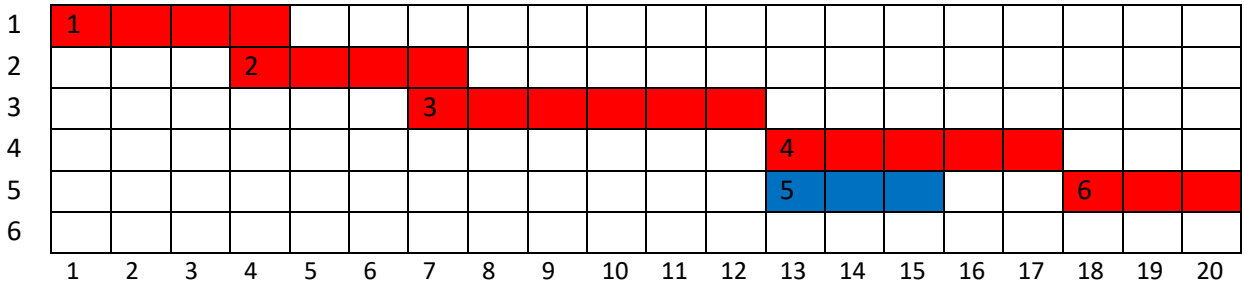
Output	Fish Length (cm)	Notes
Catch Accepted	12	Minimum valid boundary. 12 cm is accepted because it is equal to the lower limit.
Catch Accepted	40	Maximum valid boundary. 40 cm is accepted because it is equal to the upper limit.

Error – Invalid Length	11.9	Below minimum boundary. Any value less than 12 cm is rejected.
Error – Invalid Length	40.1	Above maximum boundary. Any value greater than 40 cm is rejected.
Error – Invalid Length	0	Zero length is not allowed.
Error – Invalid Length	-5	Negative values are invalid.

5. Timeline _____/10

Task ID	Task Description	New Duration (Hours)	Predecessors
1	Requirements & Design	**3**	None
2	Database Setup	**3**	1
3	Implement Record Catch	**5.5**	2
4	Implement Achievement Logic	**5.5**	3
5	View Catalogue & Achievements	**3**	3
6	Testing & Debugging	**3**	4, 5

Pert diagram



Gantt timeline